E-commerce Customer Churn Prediction Using Predictive Analytics

Project Overview

Customer churn significantly impacts revenue forecasting, marketing budgets, and overall operational growth in the e-commerce industry. This project uses predictive analytics and machine learning to identify customers who are likely to churn (stop shopping) before they actually quit the platform.

The project explores customer purchasing history, performs extensive data preprocessing and feature engineering, and develops classification models to predict churn risk accurately.

Project Objectives

- Analyze e-commerce customer transaction and session data to identify churn patterns.
- Perform Exploratory Data Analysis (EDA) to uncover key user trends.
- Clean and preprocess raw, noisy customer data.
- Engineer meaningful behavioral features to improve predictive performance.
- Build machine learning models to classify churn risk.
- Generate actionable business recommendations to support e-commerce marketing and operations.

Dataset Description

The dataset contains customer demographic profiles, digital session footprint records, and purchasing behaviors.

Key Features

Feature	Description
`Age`	Customer's age in years
`Membership _Years`	Number of years active as a member
`Login _Frequency`	Average number of monthly app/website logins
`Session _Duration _Avg`	Average time spent per session (in minutes)
`Pages _Per _Session`	Average number of product pages visited per session
`Cart _Abandonment _Rate`	Percentage of shopping carts abandoned without purchase
`Days _Since _Last _Purchas`	Number of days elapsed since the customer last completed a checkout
`Customer _Service _Calls`	Total number of support calls/chats initiated
`Lifetime _Value`	Accumulated revenue generated by the customer
`Credit _Balance`	Remaining store credit or loyalty points on the user’s account

Target Variable

Value	Meaning
0	Active (Customer has not churned)
1	Churned (Customer stopped using the service)

Project Workflow

Day 1 — Exploratory Data Analysis (EDA)

- Examined the raw customer dataset structure and data types.
- Identified missing values and flagged overall data quality issues.
- Generated descriptive statistics to understand variable distributions.
- Detected outliers in continuous fields like `Average _Order _Value` and `Session _Duration _Avg`.
- Explored visual relationships between app usage behaviors and churn rates.

Day 2 — Data Cleaning & Preprocessing

- Removed unique, high-cardinality identity columns to prevent data leakage.
- Handled missing values using target-driven column imputations.
- Removed duplicate records across the database.
- Converted categorical variables (`Gender`, `Country`, `Signup _Quarter`) into numeric indicators using One-Hot Encoding.

Day 3 — Feature Engineering

Several new features were engineered to capture interactive user behaviors:

- **Shopping Engagement Score:**

Login _Frequency_ Session * session _Duration _Avg

- **Customer Lifetime Activity:**

Total _Purchases + Product _Reviews _Written

- **High Cart Abandonment Indicator:**

1 if Cart _Abandonment _Rate > 70 % else 0

- **Inactivity Window:**

Days _Since _Last _Purchase * Customer _Service _Calls

Day 4 — Model Building & Evaluation

To avoid data leakage, all non-numeric helper targets were dropped before training. Continuous features were scaled using standard normalization fitted strictly on the training set.

Models Trained:

- Logistic Regression
- Decision Tree Classifier

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Day 5 — Insights & Recommendations

- Analyzed model performances and feature importance.
- Identified major behavioral drivers of customer churn.
- Generated operational marketing recommendations for the e-commerce business.

Model Performance

Metric	Logistic Regression	Decision Tree Classifier
Accuracy	78.34%	79.45%
Precision	69.83%	68.07%
Recall	44.15%	43.26%

Metric	Logistic Regression	Decision Tree Classifier
F1-Score	0.5410	0.5290

Best Performing Model

Decision Tree Classifier

The Decision Tree outperformed Logistic Regression across overall accuracy and showed a stronger ability to classify customer churn behavior by naturally mapping out complex, non-linear cutoffs in user inactivity.

Key Insights

- **Support Calls Warn of Attrition:**

Customers making frequent customer service calls show a major spike in churn risk due to unresolved friction.

- **Account Recency Matters:**

The longer a customer goes without checking out, the higher the chance they have switched to a competitor.

- **Loyalty Grows with Time:**

Customers with over 2 years of membership history show incredibly low churn tendencies compared to fresh sign-ups.

- **Cart Abandonment is a Warning Sign:**

A surge in cart abandonment suggests active comparison shopping, serving as a direct warning bell.

- **High Spenders Stay put:**

Customers who maintain a positive store credit balance or show high lifetime spend remain reliably active.

Business Recommendations

- **Proactive Customer Care Escalations:**

Trigger priority support outreach whenever an account crosses 3 customer service calls.

- **Automated Win-Back Flows:**

Schedule automated re-engagement email campaigns with exclusive coupons as a customer's inactivity approaches 45 days.

- **Onboarding Retention focus:**

Deploy customized welcoming campaigns during a customer's first 6 months to secure early-stage loyalty.

Limitations

- **Missing Real-World Events:**

The data doesn't see real-world events like a competitor's holiday sale or a sudden shipping delay.

- **Missing Information:**

Some customer files had blank spots, forcing us to make educated guesses during cleanup.

- **Unbalanced Data:**

There are far more active customers than leaving customers, making it harder for the model to learn churn patterns.

- **Only Two Models Tested:**

We only tried two basic models, leaving more powerful and advanced tools untouched.

- **No Customer Moods:**

The dataset only looks at raw numbers, completely missing out on text complaints, reviews, and customer emotions.

Possible Improvements

- **Try Smarter Models:**

Upgrade to advanced algorithms like Random Forest or XGBoost to catch hidden customer habits.

- **Fine-Tune the Dials:**

Adjust the internal math settings of our models to scrape out a higher final accuracy score.

- **Fix the Balance:**

Use coding tricks like SMOTE to balance the data so the model gets more practice looking at leaving customers.

- **Add Helpful Outside Info:**

Feed the model extra details like seasonal sales calendars, email click rates, or coupon histories.

- **Drop Useless Columns:**

Delete distracting variables from the data to help the computer run its numbers faster and cleaner

Tools & Technologies

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Development Environment

- Jupyter Notebook
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Future Improvements

Advanced Ensemble Models

- Random Forest
- Gradient Boosting
- XGBoost

Hyperparameter Optimization

- Grid Search
- Random Search

Additional Features

- Competitor Price Indexing (Monitoring competitor price drops and sales events)
- Seasonal Demand Indicators (Tracking high-churn seasons like post-holiday quarters)
- Customer Support Ticket Sentiment (Extracting emotional tone from support chat logs)
- Delivery Performance Metrics (Factoring in exact shipping delays or product damage rates)

Cross-Validation

Stratified K-Fold Cross Validation (Implementing a robust K-Fold split to ensure stable, reliable model evaluations across all training runs)

Repository Structure

ecommerce-customer-churn-prediction/

```
│
├── data/
├── notebooks/
├── images/
├── README.md
├── requirements.txt
└── model.pkl
```

Author

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- Passionate about transforming messy raw data into actionable business insights through predictive modeling.